

Fig. 5: An actual-size PCB layout of the transmitter circuit

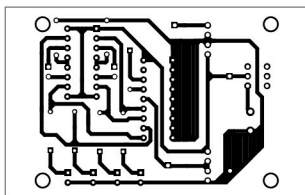


Fig. 7: An actual-size PCB layout of the receiver circuit

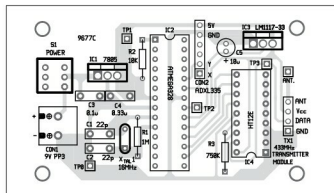


Fig. 6: Component layout of the transmitter circuit

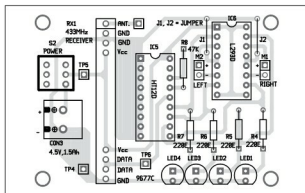


Fig. 8: Component layout of the receiver circuit

TABLE II
Test Points

Test point	Details
TP0	0V (GND)
TP1	5V
TP2	3.3V
TP3	Train of pulses
TP4	0V (GND)
TP5	4.5V
TP6	Train of pulses

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

is uploaded into the microcontroller, gesture.ino code of this project can be uploaded.

Construction and testing

An actual-size, single-side PCB layout of the transmitter circuit is shown in Fig. 5 and its component layout

in Fig. 6. An actual-size, single-side PCB layout of the receiver circuit is shown in Fig. 7 and its component layout in Fig. 8.

The transmitter section can be held in your palm or on the other side (refer Fig. 9). The receiver module is mounted on the robot.

Mount all components on the PCBs shown here to minimise assembly errors. Fix the receiver PCB and 4.5V battery on the chassis of the robot. Fix two motors, along with wheels, at the rear side of the robot and a castor wheel on the front. After uploading the main code into the microcontroller, remove it from the Arduino Uno board and insert it into the populated transmitter PCB.

Now, switch on the power supplies in the transmitter as well as receiver circuits. Attach the transmitter circuit to your hand and move your hand forwards, backwards and sideways. Directions of the robot movement are given in Table I. The robot will stop if you keep your palm hori-

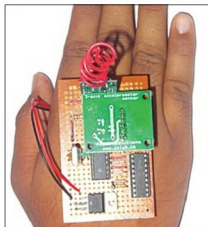


Fig. 9: Transmitter module mounted on the back of the palm

zontal, parallel to the Earth's surface.

For troubleshooting, first verify that voltages at various test points are as per Table II. ●



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